

Minh Phuc Nguyen

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EDUCATION

Union College

Schenectady, NY

Bachelor of Science in Computer Engineering

Expected graduation: June 2027

- Relevant Courses: Object-Oriented Programming, Data Structures & Algorithms, Parallel and High Performance Computing, CUDA Advanced Libraries, Computer Architecture, Embedded Systems, Machine Learning

TECHNICAL SKILLS

Programming Language: C++, CUDA, Python, Java, Bash, Makefile

GPU & Parallel Computing: Nsight Systems, CUDA Streams, OpenMP, CUB, Thrust, TensorRT

Frameworks & Libraries: PyTorch, OpenCV, NumPy, Pandas, NCCL

Tools & Platforms: Git, Valgrind, Linux Terminal, CMake

EXPERIENCE

Research Assistant - Matrix Multiplication Project

Schenectady, NY

Union College's Computer Lab

01/2024 - Present

- **Improved search algorithms** for new algebraic approaches to faster matrix multiplication using C++ and CUDA in a heterogeneous HPC setup.
- Built a GPU data pipeline across 8x NVIDIA L4 GPUs, reducing runtime by 20x compared with the CPU baseline for large-scale experiments.
- Used Nsight Systems to find GPU bottlenecks and overlap compute with data transfers, increasing GPU utilization from 30% to 95%.

GPU Software Engineer

Undisclosed Hedge Fund

- **Built a C++/CUDA financial simulation engine** with CPU-GPU correctness checks, reducing backtesting latency for risk and pricing workloads.
- Reworked GPU market-data processing with CUDA streams and OpenMP, using Nsight Systems to overlap compute and memory transfers and reach a 10x speedup over naive kernels.
- Implemented GPU math routines with CUB and Thrust for reductions and scans, improving throughput in numerical pipelines for quant models.

PROJECTS

CUDA-based Real-Time Video Stream Filtering

Schenectady, NY

Personal Project

- **Built a benchmark system** to compare GPU filtering kernels using CUB, CUDA streams, shared memory tiling, and other CUDA optimizations; the CUB version achieved 52% higher FPS.
- Improved video-processing throughput by 70%+ using pinned host memory, multi-stream execution, warp-level primitives, and Thrust.

Autonomous Driving Perception System

Schenectady, NY

Embedded Vision Project

- **Trained a semantic segmentation model** on 2x GPUs with data parallelism, overlapping gradient synchronization with backpropagation to reduce training time by 60% versus a single-GPU run.
- Optimized Jetson Orin Nano inference with TensorRT by converting the model to a static graph and fusing conv, bias, and ReLU operations, reaching 20 FPS at 1080p.

Transformer Training System

Schenectady, NY

Distributed Deep Learning Project

- **Rebuilt the Transformer** from *Attention Is All You Need* and trained it with NCCL pipeline parallelism, splitting layers across GPUs and using micro-batches to reduce pipeline stalls; improved throughput by 1.6x over a naive model-parallel baseline.
- Used torch.compile to capture the forward/backward graph and fuse attention, MLP, and normalization operations, reducing Python and kernel-launch overhead during training.